Assignmnet 8 Solution

Problem 1

We can formulate this as a binary optimization problem.
Let x_i be a binary decision variable where:

- $x_i = 1$ if player i is in the starting lineup
- $x_i = 0$ otherwise
  Then the constaints are shown as following:
  (a) $\sum_{i=1}^{7} x_i = 5$
  (b) $x_3 + x_4 + x_5 + x_6 + x_7 \geq 2$
  (c) $3x_1 + x_2 + 3x_3 + 3x_4 + x_5 + 2x_6 + 2x_7 \geq 10$
  (d) $x_3 + x_6 \leq 1$
  (e) $x_2 + x_3 \geq 1$

$x_i \in \{0, 1\}$, for i = 1,...,7

Problem 2

(a)

Let:

- x_{ij} be the number of workers assigned to product i on line j.
- y_j be a binary variable where $y_j = 1$ if line j is used, and $y_j = 0$ otherwise.
  So the objective function will be
  $$\text{Minimize } 2000y_1 + 3000y_2 + 4000y_3 + 700x_{11} + 700x_{21} + 700x_{31} + 1000x_{12} + 1000x_{22} + 1000x_{32} + 1500x_{13} + 1500x_{23} + 1500x_{33}$$

For constraints,

1. Production Requirements: Ensure each product's demand is met by the workers assigned:

- For Product 1: $50x_{11} + 90x_{12} + 120x_{13} \geq 600$
- For Product 2: $75x_{21} + 110x_{22} + 130x_{23} \geq 800$
- For Product 3: $90x_{31} + 125x_{32} + 150x_{33} \geq 1000$

2. Worker Availability: The total number of workers cannot exceed the available workforce:

- $\sum_{i=1}^{3} \sum_{j=1}^{3} x_{ij} \leq 20$

3. Line Usage: A line must be activated (set $y_j = 1$) if any workers are assigned to it, for each line j :

- $x_{1j} + x_{2j} + x_{3j} \leq 20y_j$ for each j

4. Integer and Binary Variables: All x_{ij} must be non-negative integers and all y_j must be binary:

- $x_{ij} \geq 0$ and integer for all i, j
- $y_j \in \{0, 1\}$ for all j

(b)

```
cvx_begin

cvx_solver mosek


% Define variables
variable x(3, 3) integer
variable y(3) binary


minimize (700 * sum(x(:,1)) + 1000 * sum(x(:,2)) + 1500 * sum(x(:,3)) + 2000 * y(1) + 3

subject to


    50 * x(1,1) + 90 * x(1,2) + 120 * x(1,3) >= 600;
    75 * x(2,1) + 110 * x(2,2) + 130 * x(2,3) >= 800;
    90 * x(3,1) + 125 * x(3,2) + 150 * x(3,3) >= 1000;
    sum(x(:,1))+ sum(x(:,2))+sum(x(:,3))<= 20;
    for j = 1:3
        sum(x(:, j)) <= 20 * y(j);
    end
    x >= 0;


cvx_end

% Display results

disp('Optimal solution x:')
disp(x)
```

```
Status: Solved
Optimal value (cvx_optval): +30500

Optimal solution x:
   (2,2)       5.0000
   (3,2)       8.0000
   (1,3)       5.0000
   (2,3)       2.0000
>>
```

- Line 2 should be set up
  With 5 workers producing Product 2, 8 workers producing Product 3, and no workers producing Product 1.
- Line 3 should be set up
  With 5 workers producing Product 1, 2 workers producing Product 2, and no workers producing Product 3.
- Line 1 should not be set up.
- Total number of workers is $5 + 8 + 5 + 2 = 20 \leqslant 20$
- The minimum total cost will be 30500$.

Problem 3

(a)

Define the notations:
x_1: Calculus;
x_2: Operations Research;
x_3: Data Structures
x_4: Business Statistics;
x_5: Computer Simulation;
x_6: Intro to Programming;
x_7: Forecasting.

- Objective function $\min \sum_{i=1}^{7} x_i$.
- Constraints:
  (1) at least two math courses
  $x_1 + x_2 + x_3 + x_4 + x_7 \geqslant 2$

(2) at least two OR courses

$x_2 + x_4 + x_5 + x_7 \geqslant 2$

(3) at least two computer courses.

$x_3 + x_5 + x_6 \geqslant 2.$

(4) Prerequisite 4 with 1

$x_4 \leqslant x_1$

(5) Prerequisite $3\&5$ with 6

$x_3 \leqslant x_6, x_5 \leqslant x_6.$

(6) Prerequisite 7 with 4

$x_1 \leqslant x_4$

Therefore, IP Formulation: $\min \sum_{i=1}^{7} x_i$

$$
\begin{aligned}
\text{s.t. } & x_1 + x_2 + x_2 + x_4 + x_7 \geqslant 2 \\
& x_2 + x_4 + x_5 + x_7 \geqslant 2 \\
& x_3 + x_5 + x_6 \geqslant 2 \\
& x_4 \leqslant x_1 \\
& x_3 \leqslant x_6 \\
& x_5 \leqslant x_6 \\
& x_7 \leqslant x_4 \\
& x_i = \{0, 1\}, \text{ for } \forall i = 1, 2, \ldots, 7
\end{aligned}
$$

(b)

To graduate from FastTrack University with a major in OR, a student needs to take at least 4 courses. The courses combination can be Calculus, Business Statistics, Computer Simulation, and Intro to Programming. (Operation Research, Data Struction, Computer Simulation, and Intro to Programming)

```
cvx_begin
cvx_solver Mosek

variable x(7) binary


minimize(sum(x))

subject to
    x(1) + x(2) + x(3) + x(4) + x(7) >= 2
    x(2) + x(4) + x(5) + x(7) >= 2
    x(3) + x(5) + x(6) >= 2
    x(4) <= x(1)
    x(3) <= x(6)
    x(5) <= x(6)
    x(7) <= x(4)

cvx_end


fprintf('Optimal values of variables x:\n');
disp(x);
fprintf('Objective function value: %f\n', sum(x));


Status: Solved
Optimal value (cvx_optval): +4

Optimal values of variables x:
     1
     0
     0
     1
     1
     1
     0

Objective function value: 4.000000
>>
```

Problem 4

(a)

For this problem setting, we need at least two variable equal to 1. So $x_1 = 0, x_2 = 1, x_3 = 1$ is a set of optimal solution based on inspection.

(b)

To show the constraints can be replaced, we can show that the feasible sets of the two problems are identical. The original constraints ensure that at least one of must be 1 in every pair. So the feasible set is that at least two of the variables must be 1. It is the same as the second constraints $x_1 + x_2 + x_3 \geq 2$. Thus the constraints can be replaced.

(c)

```
cvx_begin

cvx_solver mosek

variable x(3)

minimize 16*x(1)+14*x(2)+15*x(3)

subject to

    % x(1)+x(2)+x(3) >= 2
    x(1) + x(2) >= 1
    x(2) + x(3) >= 1
    x(3) + x(1) >= 1
    0 <= x(1) <= 1
    0 <= x(2) <= 1
    0 <= x(3) <= 1
cvx_end

disp("Optimal solution:")
disp(x)
```

```
------------------------------------------------------------
Status: Solved
Optimal value (cvx_optval): +22.5

Optimal solution:
    0.5000
    0.5000
    0.5000


cvx_begin

cvx_solver mosek

variable x(3)

minimize 16*x(1)+14*x(2)+15*x(3)

subject to

    x(1)+x(2)+x(3) >= 2
    % x(1) + x(2) >= 1
    % x(2) + x(3) >= 1
    % x(3) + x(1) >= 1
    0 <= x(1) <= 1
    0 <= x(2) <= 1
    0 <= x(3) <= 1
cvx_end

disp("Optimal solution:")
disp(x)
```

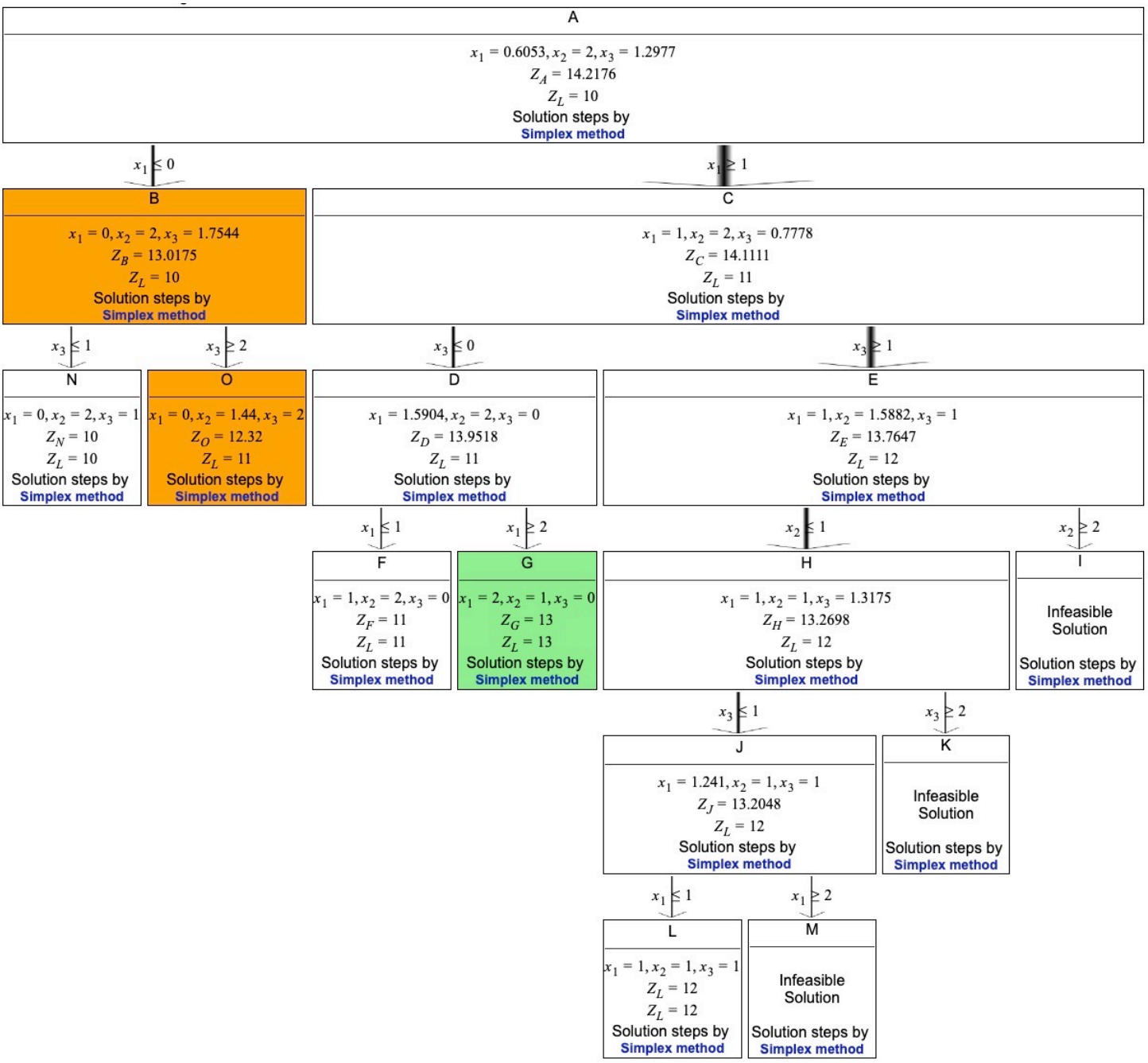
```
-------------------------------------------------------------
Status: Solved
Optimal value (cvx_optval): +29

Optimal solution:
     0
     1
     1

>>
```

Using the second constraints, we can get the integral optimal solution. So using the second constraints is better.

Problem 5



Notice that all the coeffients in the objective function are integral. So we don't necessarily need to branch B, E since we can not get better integral solution greater than 13.

```
cvx_begin

    cvx_solver Mosek



    variable x1
    variable x2
    variable x3

    % variable x1 integer
    % variable x2 integer
    % variable x3 integer

    0 <= x1 <= 2;
    0 <= x2 <= 2;
    0 <= x3 <= 2;


    maximize(5*x1 + 3*x2 + 4*x3)

    % Constraints
    subject to
        43*x1 + 25*x2 + 57*x3 <= 150
        83*x1 + 34*x2 + 63*x3 <= 200
cvx_end

% Display results
fprintf('Optimal values:\n');
fprintf('x1 = %d\n', x1);
fprintf('x2 = %d\n', x2);
fprintf('x3 = %d\n', x3);
fprintf('Maximum objective value: %f\n', cvx_optval);
```